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Name:

Seat #:

MIDTERM EXAMINATION

ECON 201, Spring 2014
Intermediate Macroeconomics
Prof. Ryan Edwards

March 10, 2012

RULES:

- Don't start until I say so.
- You may check that you have all 10 pages.
- Write your name and **seat number** at the top *and on page 7*.
- When I call time, **stop**. If you don't, you get an F.
- No books, no notes, no calculators, **no electronic devices**.
- This exam has 1 whimsical warmup and 4 main questions together worth 75 points, one point for each minute of time you have. Budget your time.
- If I catch you cheating in any way, you will receive an F for the course and a referral to the VP for Student Affairs.

BE GOOD, NEVER BE EVIL

Points:	1	5	2	3	4	5	Total:
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1. (5 points total) *Whimsical Warmup.*

THERE ARE NO WRONG ANSWERS. You will get all 5 points even if you don't answer anything. See the large "5" above?

- (a) (1 point) Bruno Mars performed the Super Bowl Halftime Show this year. What did you think?
 - i. Sally Venus would have been better
 - ii. Mmmmm, Mars bar
 - iii. Sweet
 - iv. Kind of dull but also intergenerationally appealing. I just really like that nice young man

- (b) (1 point) The Denver Broncos:
 - i. Splat
 - ii. Boom!
 - iii. Who?
 - iv. Yes.

- (c) (1 point) Miley Cyrus and Justin Bieber:
 - i. Weigh about 100 lbs *combined*
 - ii. Don't tweet enough, really
 - iii. One is jealous of the other's last name. "Sigh" just isn't a cool verb like "Believe"
 - iv. Two great things that would go great together, but oddly they haven't

- (d) (1 point) The food on campus at Queens College
 - i. Is super yummy
 - ii. There's food on campus?
 - iii. Is so amazing *because* of all the off-campus competition from great local restaurants
 - iv. Technically is all off-campus in food trucks. Thankfully

- (e) (1 point) Econ 201 is
 - i. Burp
 - ii. My BFF of econ courses. I heart you, Econ 201
 - iii. Something that keeps me awake at night
 - iv. Thankfully half over

2. (10 points total) *Short Answer Potpourri.*

(a) (2 points) Write a sentence describing what “capital” is in Econ 201.

(b) (2 points) Write a sentence describing what this equation means in Econ 201, using words like “production” or “income.”

$$Y = AK^{1/3}L^{2/3}$$

(c) (2 points) Recall that nominal GDP (Y^N) is related to real GDP (Y^R) and the price level (P) in the following way:

$$Y^N = P Y^R$$

If $g[P] = 0.03$ and $g[Y^R] = 0.02$, what is $g[Y^N]$?

(d) (2 points) If I deposited \$100 into a bank account that paid 2% interest annually, how much would I have accumulated after 2 years? You do not need a calculator, just write down a mathematical expression.

(e) (2 points) The marginal product of labor is given by

$$MPL = \frac{2}{3}A \left(\frac{K}{L} \right)^{1/3}$$

If technology (A) increases by 5%, what happens to the wage, if anything? Be specific.

3. (25 points total) *Famine and out-migration in the Solow Model.*

St. Patrick's Day 2014 is next Monday. In 2011, 34.5 million U.S. residents claimed Irish ancestry, meaning there were 7 Irish Americans for every single resident of Ireland today, which had a population of 4.68 million.¹ What events could have led to such a situation?

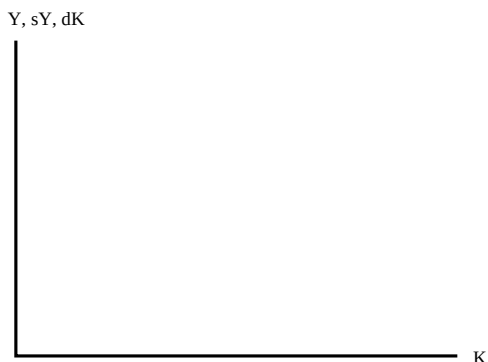
The Great Famine of 1846–47 in Ireland was caused by a massive failure of the potato crop due to infestation by a *potato blight*, a disease that kills potatoes. How can we view this event and its implications from the perspective of the Solow Model?

Recall that the key equations of the Solow Model are:

$$Y = A K^{1/3} L^{2/3} \quad (1)$$

$$\Delta K = sY - dK \quad (2)$$

- (a) (4 points) On the axes below, draw the Solow diagram for 1845 Ireland, identifying the saving curve, depreciation curve, income curve, and the steady state.

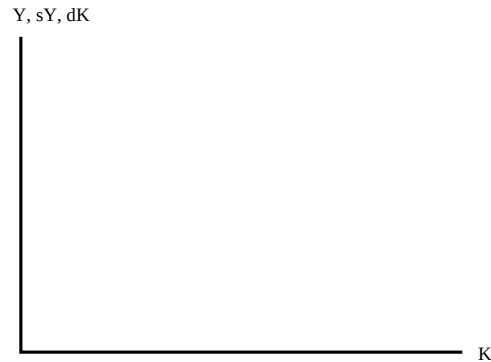


Unlike a plague or virus that kills people, the potato blight kills potatoes. Even with the same amount of capital available, potato farmers were vastly less productive after the blight arrived because the fruits of their labor simply decayed away.

- (b) (3 points) What parameter in the Solow Model can we use to model the effects of the potato blight? What does the blight do to that parameter? Explain.

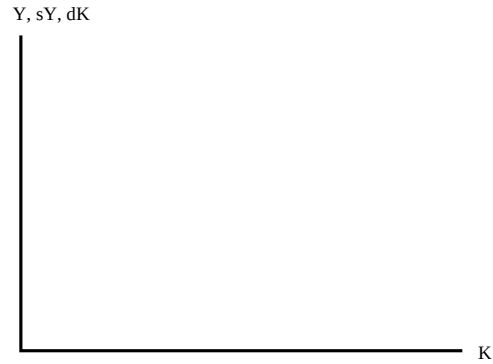
¹http://www.census.gov/newsroom/releases/pdf/cb13-ff03_st_patricks_day.pdf

- (c) (4 points) Redraw the Solow diagram depicting the change you described in part (b) for 1846 Ireland, identifying the saving curve(s), depreciation curve(s), income curve(s), and the steady state(s). If they are different, show “before” and “after” versions as appropriate, and briefly explain what you have drawn.



- (d) (3 points) What will happen in the short run to income per person (Y/L) as a consequence of what you have shown in part (c)? What about in the long run? Explain.
- (e) (3 points) How would you advise an Irish resident who was facing what you described in part (d)? Would you suggest that he or she migrate to the United States, or stay put in Ireland? Remember that crossing the Atlantic Ocean was very costly in terms of time and money in 1846. Explain fully.

- (f) (4 points) Supposing that a large number of Irish residents took the advice you gave in part (e), illustrate the joint effects of that change along with the original change you showed in part (c), and explain. Discuss the net effect on income (Y).



In the Solow Model, income per person in the short run equals

$$\frac{Y}{L} = A \left(\frac{K}{L} \right)^{1/3}$$

while in the long run, in the steady state, it equals

$$\frac{Y^*}{L} = A^{1/2} \left(\frac{s}{d} \right)^{1/2}$$

- (g) (4 points) Discuss the short-run and long-run effects of the development you depicted above in part (f) on income per Irish person (Y/L). What are they? Are they the same, or are they different? Explain and discuss using some intuition.

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4. (10 points) *A Model Immigrant*

Recall that in the simple neoclassical production model of chapter 4, the marginal product of labor is given by:

$$MPL = \frac{2}{3}A \left(\frac{K}{L} \right)^{1/3} \quad (3)$$

- (a) (2 points) Qualitatively, what is the first-order effect of the arrival of more immigrants who arrive to work? That is, what does immigration change, if anything, in the simple model of production?

- (b) (4 points) On the axes below, draw the labor demand curve(s), vertical labor supply curve(s), and show the market-clearing wage(s) in the labor market both before and after a number of immigrants arrive.



- (c) (2 points) If immigrants bring capital with them or save and produce capital after they arrive, how might that change the story, if at all? Explain, with references to the graph.
- (d) (2 points) If immigrants save after they arrive and send their savings back to family residing in their countries of origin, how might that change the story, if at all? Explain, with references to the graph.

Points:	1	5	2	3	4	5	Total:
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5. (25 points) *Government, R&D, and the Romer Model.*

Does a Romer economy occur naturally, through market forces? Put another way, if all factor are paid their marginal products, will sustainable growth occur naturally?

Recall that the Romer model includes these equations:

$$\Delta A_t = \bar{z} A_t L_{at} \quad (4)$$

$$Y_t = A_t L_{yt} \quad (5)$$

$$\bar{N} = L_{at} + L_{yt} \quad (6)$$

$$L_{at} = \bar{\ell} \bar{N} \quad (7)$$

For simplicity, let's call L_{yt} "farmers" and L_{at} "scientists."

Part I: Solving the standard Romer Model.

(On this page, you may "skip to the end" and not follow the steps as I have laid them out, **if** you show your work.)

- (a) (2 points) Use equations (4) and (7) to find an equation for the growth rate of A in terms of $\bar{z}$, $\bar{\ell}$, and $\bar{N}$.

- (b) (2 points) Use equations (5), (6), and (7) to find an equation for income per person, $Y/\bar{N}$, in terms of A_t and $\bar{\ell}$.

- (c) (2 points) Use your answer to part (a) to find an equation for A_t using its starting level A_0 , and then substitute for A_t in your answer to part (b) in order to find an equation for income per person, $Y/\bar{N}$, solely in terms of the parameters: A_0 , $\bar{z}$, $\bar{\ell}$, and $\bar{N}$.

Part II: Interpreting the Romer Model.

The marginal product of farmers, L_{yt} , is equal to A_t . That is also their average product because there are constant returns to labor in goods production in this standard version of the Romer Model.

- (d) (2 points) If farmers, L_{yt} , are paid their marginal product as their wage, w_y , write down a formula for the total wages paid to labor in goods production.

- (e) (2 points) Subtract your answer to part (d) from total goods production given by equation (5), which should equal the total amount of output that is paid to other factors of production. Briefly explain what you have found.

In addition to farmers, L_{yt} , there are scientists, L_{at} . *The marginal product of scientists, in terms of new ideas, is equal to $\bar{z}A_t$.* That is, competitive markets would pay scientists in ideas.

- (f) (2 points) Can scientists eat their ideas? What would they have to do with their ideas in order to survive? Explain.

- (g) (2 points) Scientists are paid in ideas, but do they really “own” those ideas they are paid? Can they trade them for other things? Explain.

(h) (3 points) Think about what you have found. If competitive markets determine the allocation of labor, to what level would the scientist share $\bar{\ell}$ quickly converge? Why?

(i) (3 points) From the perspective of a “newborn” individual deciding to become either a farmer or a scientist, what would have to be true about the levels of the wages in those two fields to make that individual stick with his or her occupational choice rather than switch?

Imagine government levied a tax rate τ on farmers, so that their after-tax wage was $w'_y = (1 - \tau)A_t$ and that total tax collections were $\tau A_t L_{yt}$. Suppose the government divided these collections equally among the L_{at} scientists, so that scientists earned a wage equal to

$$w'_a = \frac{\tau A_t L_{yt}}{L_{at}} \quad (8)$$

(j) (5 points) What tax rate τ would support a scientist share of $\bar{\ell}$? [Hint: set $w'_a = w'_y$ and solve, using equation (6) to substitute for L_{yt} and L_{at} in terms of $\bar{\ell}$ and $\bar{N}$.]